

Multi-Task Deep Learning Framework for Unified Cattle Health and Behavior Monitoring

1st Hasin Ishrak
Dept. of Computer Science
and Engineering
BRAC University
Dhaka, Bangladesh
hasin.ishrak@g.bracu.ac.bd

2nd Namira Abrar Haque
Dept. of Computer Science
and Engineering
BRAC University
Dhaka, Bangladesh
namira.abrar.haque@g.bracu.ac.bd

3rd Sanjida Akter Bithi
Dept. of Computer Science
and Engineering
BRAC University
Dhaka, Bangladesh
sanjida.akter.bithi@g.bracu.ac.bd

4th Shouvik Banik
Dept. of Computer Science
and Engineering
BRAC University
Dhaka, Bangladesh
shouvik.banik@g.bracu.ac.bd

5th Nusrat Lamya Faruk
Dept. of Computer Science
and Engineering
BRAC University
Dhaka, Bangladesh
nusrat.lamya.faruk@g.bracu.ac.bd

Abstract—Automated computer vision in precision livestock farming increasingly targets distinct physiological and behavioral traits, including Body Condition Scoring (BCS), behavior recognition, and individual cow re-identification (Re-ID). These tasks demand distinct visual representations, including static body morphology, temporal posture dynamics, and biometric coat pattern appearance. In this paper, we evaluate task-specific cattle-centered visual representations—incorporating target localization, binary foreground masks, and lightweight temporal modeling—across leakage-aware single-task and multi-task settings over a shared ResNet-18 backbone. We evaluate three multi-task paradigms against matched single-task references: (i) monolithic hard parameter sharing (E1), (ii) modular task-private residual bottleneck adapters (E3), and (iii) gradient conflict projection via PCGrad (E4). Across matched evaluation cohorts (7,549 burst-group-disjoint ScienceDB images, 780 temporal behavior sequences from CVB and Kaggle Beef, and 69 held-out cows evaluated against a 36,811-image gallery under SideView Protocol A), cattle-centered representations provide task-dependent benefits, lowering BCS MAE from 0.1929 to 0.1709 and lifting Snapshot Re-ID Rank-1 from 38.88% to 62.93% under oracle mask guidance, while benefits across sub-metrics remain non-uniform. Joint hard sharing (E1) induces outcome-level negative transfer across all three tasks. Task-private adapters (E3, +3.32% parameters) fail to provide general mitigation. During E4 training, direct diagnostics record 44,177 gradient conflict projections across 16,140 super-steps (46.5%–47.8% pairwise conflict frequency). PCGrad provides selective mitigation, notably on Behavior where CVB surveillance accuracy improves (+4.26 pp over E1) and minority Walking F1 recovers (0.0408 to 0.0909), though dedicated single-task models remain superior on several primary metrics. Our framework establishes an empirical benchmark demonstrating that multi-task livestock monitoring requires balancing shared representations, task-private capacity, and conflict-aware optimization.

Index Terms—Precision livestock farming, Multi-task learning, Body condition scoring, Cattle behavior recognition, Cattle re-identification, Gradient conflict, Negative transfer.

I. INTRODUCTION

Precision livestock farming (PLF) increasingly relies on automated computer vision to monitor animal welfare, health, and productivity in dairy and beef operations [1], [2]. Three visual phenotyping tasks are central to herd management: (1) *Body Condition Scoring (BCS)*, which assesses subcutaneous fat reserves to manage nutritional balance and prevent metabolic disorders [3]–[5]; (2) *Behavior Recognition*, which tracks feeding, drinking, standing, lying, and locomotion dynamics to detect early lameness, mastitis, or heat onset [6], [7]; and (3) *Cow Re-Identification (Re-ID)*, which enables non-invasive individual traceability across facility camera views without physical tags [8], [9].

Existing systems overwhelmingly execute these tasks through separate, isolated deep neural networks [10], [11]. Maintaining independent single-task vision models presents several practical limitations: (1) *Excessive Parameter Footprint*: Executing separate convolutional backbones for condition scoring, video action recognition, and biometric retrieval triples model storage demands (34.7M parameters across three independent ResNet-18 networks), increasing hardware overhead on rural edge nodes; (2) *Susceptibility to Background Shortcut Exploitation*: Standard uncropped RGB models risk exploiting non-target background visual features—such as pen manure accumulation, chute metalwork, stall fixtures, or static shadows—rather than invariant anatomical landmarks [12], [13]; and (3) *Heterogeneous Demands and Negative Transfer*: Although visual features governing identification and condition assessment share low-level visual representations, naive parameter sharing across heterogeneous spatial and temporal objectives risks gradient conflict and negative transfer [14], [15].

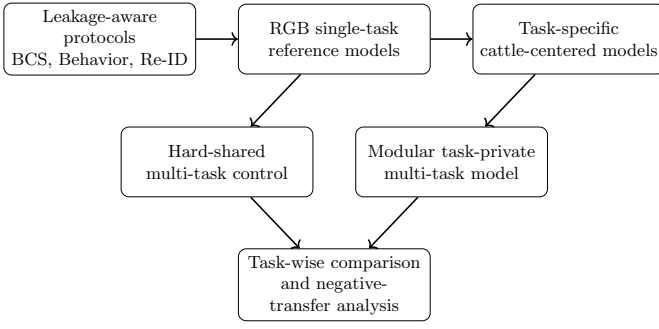


Fig. 1. Comparative research framework illustrating leakage-aware protocols, single-task baseline references, task-specific cattle-centered representations, and multi-task parameter integration strategies (E1 hard sharing, E3 modular adapters, and E4 PCGrad optimization).

To address these challenges, we evaluate a multi-task deep learning framework that concurrently assesses body condition scores, 5-class temporal behavior categories, and open-set biometric cow retrieval over a shared 4-channel convolutional backbone (Fig. 1). We examine whether cattle-centered spatial representations (combining localized cropping and foreground masking) improve individual task outcomes, and evaluate whether modular task-private adapters or gradient surgery can mitigate inter-task negative transfer.

The core contributions of this work are: (1) **Leakage-Aware Protocols**: Established verifiable benchmark protocols for three heterogeneous cattle monitoring domains: a burst-group-disjoint protocol for ScienceDB BCS; a session- and source-disjoint protocol for CVB and Kaggle Beef Behavior; and an identity-disjoint open-set retrieval protocol (SideView Protocol A) evaluating 69 held-out cows against a 36,811-image parlor gallery. (2) **Controlled Single-Task Empirical Baselines**: Provided single-task comparisons between generic RGB inputs and cattle-centered representations across matched test populations, establishing that localized cropping and foreground masks reduce BCS MAE from 0.1929 to 0.1709 and raise Snapshot Re-ID Rank-1 from 38.88% to 62.93% under oracle masks. (3) **Task-Specific Representation Strategy**: Formulated and evaluated task-tailored representations: static crops with binary masks for BCS morphology, 8-frame sequences with a lightweight Conv1D temporal network for behavior dynamics, and target-centered cropping with oracle masks for re-identification surface patterns, while empirically assessing boundaries of pose and viewpoint conditioning. (4) **Benchmark for Systematic Multi-Task Sharing**: Conducted controlled experiments comparing monolithic hard parameter sharing (E1), modular task-private residual adapters (E3), and gradient-projected optimization (E4) against matched single-task baselines on identical held-out test sets, reducing total parameter footprint by 65.7% (11.9M vs. 34.7M). (5) **Direct Shared-Backbone Gradient Diagnostics**: Recorded direct empirical measurements of gradient conflict on the shared backbone during multi-task training via PCGrad, capturing 44,177 pairwise conflict projections across 16,140 super-steps (46.5%–47.8% pairwise frequency). (6) **Calibrated Insights**

Into Multi-Task Trade-Offs: Provided empirical evidence that joint sharing induces outcome-level negative transfer across heterogeneous livestock tasks, and that architectural and optimization interventions yield selective, metric-dependent trade-offs rather than uniform improvements.

II. RELATED WORK

1) *Body Condition Scoring (BCS)*: Dairy BCS assessment traditionally relies on visual and tactile evaluation by trained assessors on 5-point ordinal scales [3], [16]. Early automated methods explored 3D point clouds or depth imaging to extract angular depressions across the cow’s hook bones, pin bones, and thurl cavity [17]. Recent works transitioned to 2D deep learning: Liu et al. [4] introduced lightweight attention networks focused on tailhead contours on rear chute images, while Yao et al. [5] evaluated side-view scoring using convolutional networks. However, standard models frequently formulate BCS as unconstrained continuous regression or multiclass classification, ignoring ordinal class structure. While ordinal deep learning formulations such as CORAL [18] enforce shared rank consistency, our BCS head is trained with independent ordinal binary cross-entropy over four cumulative threshold logits, predicting discrete score boundaries directly.

2) *Cattle Behavior Recognition*: Video-based livestock action recognition typically targets feeding, drinking, standing, and lying postures [1]. Early approaches utilized optical flow or 3D CNNs (e.g., SlowFast, I3D), which demand substantial compute and degrade under camera vibration [19]. Recent open benchmarks, including CVB [7] and CBVD [6], facilitate 2D spatial backbone combinations with Temporal Convolutional Networks (TCN) [20] or Recurrent Units. In practice, models operating on uncropped surveillance video risk developing shortcut dependence on static pen or pasture backgrounds [12], though environmental context can also provide informative scene cues.

3) *Individual Cattle Re-Identification*: Individual cow identification without physical RFID tags or neck transponders is essential for non-contact traceability. Andrew et al. [8], [21] demonstrated that coat pattern contours of Holstein-Friesian cattle serve as biometric signatures. Modern open-set metric learning architectures employ deep embeddings trained with cross-entropy and cosine-distance ranking [9]. However, domain shifts between indoor milking parlors, open barns, and pastures cause substantial retrieval degradation due to illumination, viewpoint, and posture disparities.

4) *Multi-Task Learning and Gradient Balancing*: Multi-task learning (MTL) aims to share inductive bias across related visual domains, improving memory efficiency and regularization [14]. Despite theoretical advantages, naive hard parameter sharing often exhibits *negative transfer*, where competing task objectives degrade individual task accuracies below their single-task counterparts [15]. Architectural remedies include task-private adapters, cross-stitch networks, and attention routing [22], [23]. Optimization remedies focus on gradient manipulation: GradNorm dynamically reweights loss scales based on training rates [24], while PCGrad (Projecting Conflicting

Gradients) projects task gradients onto each other's normal planes when directional conflict ($\mathbf{g}_i^\top \mathbf{g}_j < 0$) is detected [25]. In this paper, we systematically evaluate hard sharing (E1), modular bottleneck adapters (E3), and PCGrad gradient projection (E4) across multi-modal livestock tasks.

III. PROPOSED METHODOLOGY

A. Task-Specific Cattle-Centered Perception Pipelines

Because the three monitoring domains differ in visual and temporal requirements, the cattle-centered models focus on the target cow using task-specific preprocessing rather than an identical automated pipeline (Fig. 2):

- 1) **Body Condition Scoring (ScienceDB):** RT-DETR-L [26] detects cattle using the COCO cattle class with a confidence threshold of 0.25. If multiple cattle are detected, the largest bounding box is selected, and confidence breaks ties. The selected box is expanded by approximately 5% in each direction, clipped to the image boundary, and used for both the RGB crop and the SAM 2.1 prompt [27]. SAM 2.1 generates a binary mask in $\{0, 1\}$. Both RGB and mask are resized to 224×224 . RGB channels use ImageNet normalization, while the fourth channel remains an unnormalized binary mask. Perception failures (detector misses or empty masks) are excluded; no artificial replacement input is used.
- 2) **Behavior Recognition (CVB + Beef):** For CVB video clips, target-tracklet bounding boxes provided by the dataset at each sampled frame prompt SAM 2.1, and the identical box coordinates are applied to RGB and mask crops. For single-cow Kaggle Beef clips, RT-DETR-L provides the animal bounding box when available, followed by a positive center prompt to SAM 2.1; if the detector finds no cow, an image-center prompt is used. A sequence is excluded if any required mask is empty or invalid.
- 3) **Cattle Re-Identification (SideView Protocol A):** Side-View Re-ID experiments utilize the released ground-truth target-cow segmentation masks, operating strictly as an *oracle segmentation-guided experiment*. The non-zero region of the ground-truth mask defines the bounding box, which is expanded by approximately 5%. The same crop coordinates are applied to both RGB and mask, resizing both to 224×224 . The binary fourth channel is concatenated with ImageNet-normalized RGB; RGB pixels are not multiplied by the mask.

B. Shared Spatial Trunk and Task-Specific Heads

The shared visual representation is parameterized by a 4-channel ResNet-18 backbone (11,179,648 parameters) [28]. The first convolutional layer is expanded from $\mathbb{R}^{64 \times 3 \times 7 \times 7}$ to $\mathbb{R}^{64 \times 4 \times 7 \times 7}$, with the new mask-channel weights initialized to the mean of the three pretrained RGB filters. For an input image, the trunk yields a 512-dimensional spatial feature vector $\mathbf{h} \in \mathbb{R}^{512}$.

1) **BCS Ordinal Regression Head:** ScienceDB BCS targets comprise five ordered discrete classes: 3.25, 3.50, 3.75, 4.00,

and 4.25. For a class index $y \in \{0, 1, 2, 3, 4\}$, four independent cumulative threshold targets are defined: $t_k = \mathbb{I}(y > k)$, $k \in \{0, 1, 2, 3\}$. The BCS linear head (2,052 parameters) produces four threshold logits $g_k = \mathbf{w}_k^\top \mathbf{h} + b_k$. The loss is independent ordinal binary cross-entropy:

$$L_{\text{BCS}} = -\frac{1}{4B} \sum_{i=1}^B \sum_{k=0}^3 [t_{ik} \log \sigma(g_{ik}) + (1 - t_{ik}) \log(1 - \sigma(g_{ik}))]. \quad (1)$$

During inference, the discrete class index is determined by counting threshold probabilities exceeding 0.5, then mapped back to the physical BCS scale: $\hat{y} = \sum_{k=0}^3 \mathbb{I}(\sigma(g_k) > 0.5)$, $\widehat{\text{BCS}} = 3.25 + 0.25\hat{y}$.

2) **Temporal Behavior Recognition Head:** For video clips, $T = 8$ approximately evenly spaced frames are extracted per sample, producing input tensor $[B, 8, 4, 224, 224]$. The shared 4-channel trunk processes each frame independently, yielding temporal feature sequence $\mathbf{H} \in \mathbb{R}^{B \times 8 \times 512}$. A lightweight residual Conv1D model over these frame features performs sequence modeling: the first block applies a width-three Conv1D projection from 512 to 256 channels, BatchNorm, GELU, dropout (0.2), and a width-one residual projection; the second block maintains 256 channels with BatchNorm, GELU, dropout (0.2), and an identity residual. Temporal adaptive average pooling produces a 256-dimensional feature routed to a 5-class linear classifier (723,973 parameters in head; 11,903,621 total). The head is optimized via categorical cross-entropy L_{Beh} .

3) **Biometric Re-Identification Head:** During representation learning on the 41 parlor training cows, the raw 512-dimensional feature $\mathbf{h}$ is passed to a 41-class linear classifier with bias (21,033 parameters) optimized with categorical cross-entropy loss L_{ReID} . For open-set retrieval evaluation across unseen cows, the retrieval embedding is formed by L_2 unit normalization: $\mathbf{z} = \mathbf{h} / \|\mathbf{h}\|_2$. The retrieval embedding remains 512-dimensional. Cosine similarity between normalized query and gallery embeddings ranks candidate identities.

C. Multi-Task Architectural Paradigms

1) **Monolithic Hard-Shared Baseline (E1):** As depicted in Fig. 3 (left), all three task heads branch directly from the single shared 4-channel ResNet-18 trunk (11,179,648 shared parameters; 93.74% of total capacity). Total trainable parameters equal 11,926,706. Shared trunk parameters θ_{shared} are updated via direct summation: $\mathbf{g}_{\text{shared}} = \mathbf{g}_{\text{BCS}} + \mathbf{g}_{\text{Beh}} + \mathbf{g}_{\text{ReID}}$.

2) **Modular Task-Private Residual Adapters (E3):** To evaluate whether task-private capacity mitigates negative transfer, E3 inserts an independent residual bottleneck adapter ($512 \rightarrow 128 \rightarrow 512$) between the shared feature $\mathbf{h}_{\text{shared}}$ and each task head (Fig. 3, right). Each adapter contains a linear down-projection $\mathbf{W}_{\text{down}} \in \mathbb{R}^{128 \times 512}$, Layer Normalization, GELU activation, dropout ($p = 0.1$), and a linear up-projection $\mathbf{W}_{\text{up}} \in \mathbb{R}^{512 \times 128}$. The adapter output $\Delta \mathbf{h}_t$ is added residually: $\mathbf{h}_t = \mathbf{h}_{\text{shared}} + \Delta \mathbf{h}_t$. In Behavior, the adapter operates frame-wise across all 8 frame embeddings before temporal

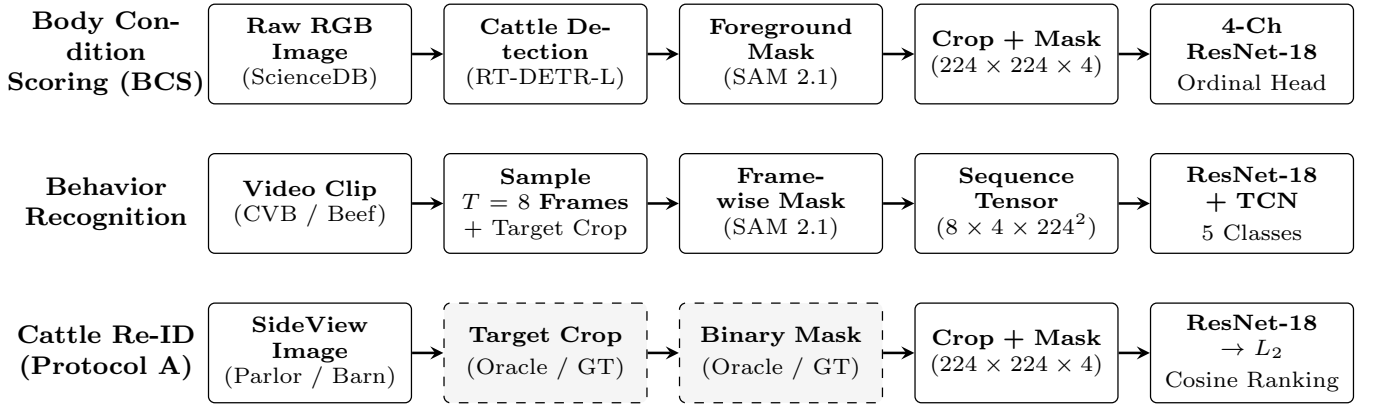


Fig. 2. Task-specific cattle-centered input representation pipelines across Body Condition Scoring (ScienceDB), Behavior Recognition (CVB / Beef), and Cattle Re-Identification (SideView Protocol A). Target localization, bounding-box cropping, and binary foreground masks generate standardized 4-channel input tensors ($224 \times 224 \times 4$), with Re-ID evaluated under released oracle ground-truth masks.

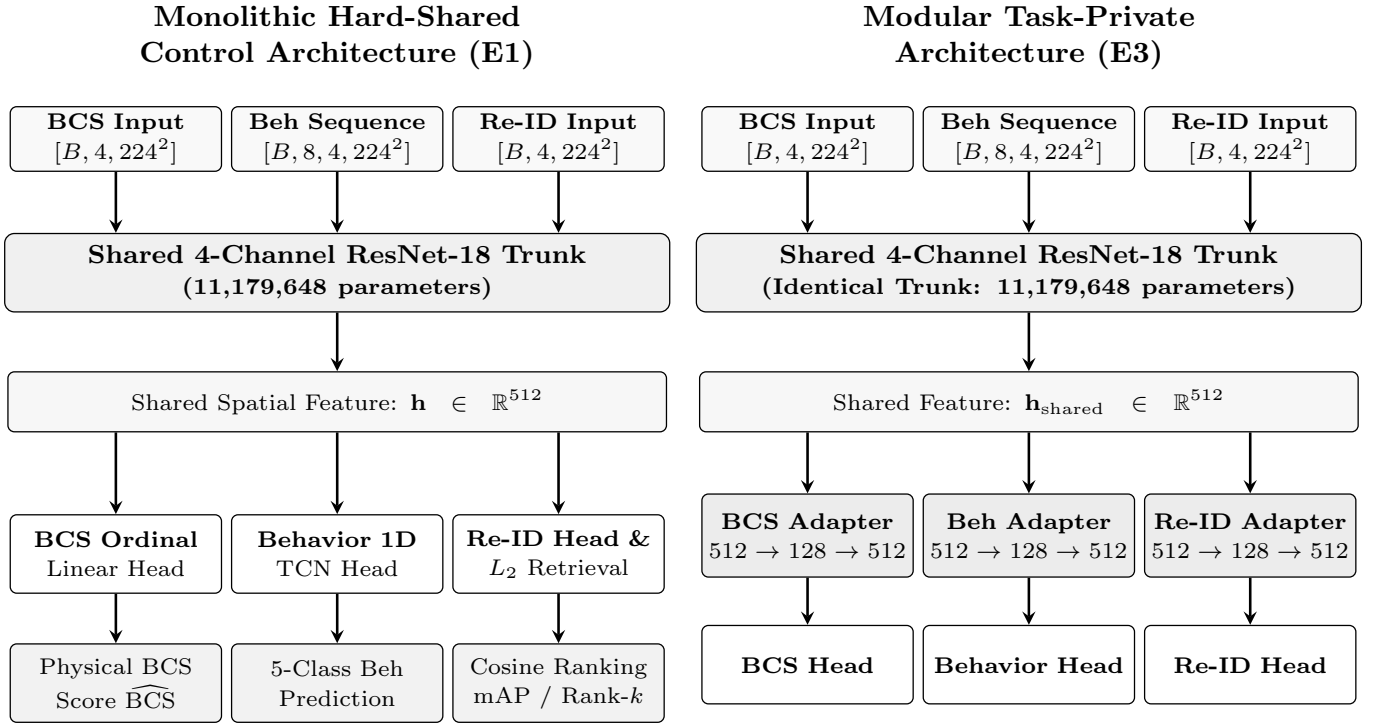


Fig. 3. Comparison of multi-task learning architectures. Left: Monolithic hard-shared control (E1) routing a single 512-dim trunk feature to all three heads. Right: Modular task-private architecture (E3) inserting bottleneck residual adapters ($512 \rightarrow 128 \rightarrow 512$) to decouple task-specific capacity.

convolution. The up-projection is initialized to zero ($\mathbf{W}_{\text{up}} = \mathbf{0}, \mathbf{b}_{\text{up}} = \mathbf{0}$), ensuring identity initialization ($\Delta \mathbf{h}_t = \mathbf{0}$) at step zero. Each adapter adds 131,968 parameters (395,904 total; +3.32% capacity), yielding 12,322,610 trainable parameters.

3) *Gradient-Projected Optimization via PCGrad (E4)*: E4 evaluates optimization-level conflict resolution while preserving the exact parameterization of E1 (11,926,706 trainable parameters; 0 adapters, 0 gates). When a pair of task gradients on the shared trunk conflicts ($\mathbf{g}_i^\top \mathbf{g}_j < 0$), $\mathbf{g}_i$ is projected onto

the normal plane of $\mathbf{g}_j$ [25]:

$$\mathbf{g}_i \leftarrow \mathbf{g}_i - \frac{\mathbf{g}_i^\top \mathbf{g}_j}{\|\mathbf{g}_j\|_2^2 + \epsilon} \mathbf{g}_j. \quad (2)$$

Pairwise projection order is deterministically shuffled at each step. Modified shared gradients are summed ($\mathbf{g}_{\text{shared}} = \sum_t \mathbf{g}_t'$). Task-head gradients remain unprojected.

D. Super-Step Multi-Task Training Protocol

Because commercial cattle datasets do not share overlapping annotations across condition, behavior, and identity, joint

Execution Sequence of One Multi-Task Training Super-Step

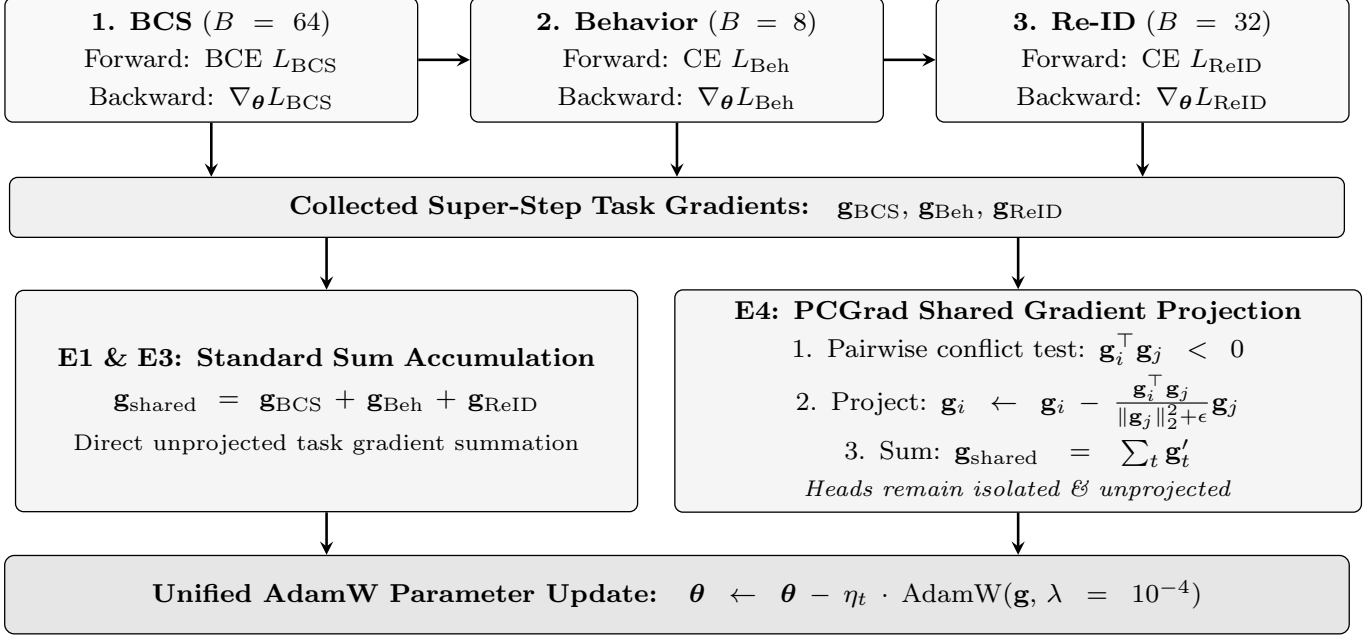


Fig. 4. Execution sequence and gradient routing of one multi-task training super-step. Sequential micro-batch gradients are accumulated; in E4 (PCGrad), conflicting shared trunk gradients ($\mathbf{g}_i^\top \mathbf{g}_j < 0$) are projected onto normal planes before the unified AdamW update.

training employs interleaved mini-batch super-steps (Fig. 4): (1) BCS forward/backward ($B_{\text{BCS}} = 64$); (2) Behavior forward/backward ($B_{\text{Beh}} = 8$ video sequences / 64 frames); (3) Re-ID forward/backward ($B_{\text{ReID}} = 32$). Task loss weights are equal ($w_{\text{BCS}} = w_{\text{Beh}} = w_{\text{ReID}} = 1.0$).

Training executes for 30 epochs with 538 super-steps per epoch (16,140 total super-steps), determined by the complete ScienceDB training population ($\lceil 34,369/64 \rceil = 538$). Optimization uses AdamW (initial learning rate 10^{-4} , cosine decay to 10^{-6} , weight decay 10^{-4} , seed 2026). Checkpoint selection is strictly determined by minimizing the composite validation objective: $L_{\text{MTL, val}} = \frac{1}{3}(L_{\text{BCS, val}} + L_{\text{Beh, val}} + L_{\text{ReID, val}})$.

IV. EXPERIMENTAL SETUP AND BENCHMARK DATASETS

A. Benchmark Datasets and Leakage Prevention

Rigorous validation in livestock vision requires preventing identity and background leakage across splits: 1) *BCS (ScienceDB)*: ScienceDB contains 53,566 RGB images across five discrete scores (3.25 to 4.25) [29]. Verified biological cow IDs are not available. To prevent near-identical video frame leakage, we enforce a repaired burst-group-disjoint protocol across 5,653 canonical burst groups: train (37,045 images; 3,958 bursts), validation (8,481 images; 850 bursts), and test (8,040 images; 845 bursts). The matched perception-successful test evaluation cohort covers 7,549 images (93.89% coverage; 489 detection misses and 2 empty masks excluded).

2) *Behavior Recognition (CVB + Beef)*: Combines CVB open-pasture surveillance video (66 source videos) [7] and

Kaggle Beef feedlot clips (201 recording sessions) [30], comprising 267 protected groups and 5,274 samples. Canonical split: train (3,785), validation (680), test (809). Walking occurs exclusively in CVB. Matched test comparison evaluates 780 retained sequences (422 CVB, 358 Beef; 29 stanchion occlusions excluded). Evaluation is source/session-group-disjoint, not cow-disjoint.

3) *Cattle Re-ID (SideViewCows2026)*: Contains 80,260 images and 80,260 masks across 110 cows [31]. Under Protocol A, 41 cows are used for representation learning (12,753 train, 2,683 val in parlor), while 69 held-out cows form the evaluation set. A fixed parlor gallery of 36,811 images is evaluated against 25,260 barn queries and 607 snapshot queries (covering 63 cows). Evaluations use released ground-truth masks (oracle benchmark).

B. Implementation Details

Models were trained using PyTorch on NVIDIA L40S GPUs. All single-task and multi-task models are evaluated from single validation-selected checkpoints (point estimates). Held-out test sets remained fully isolated from training, hyperparameter tuning, and threshold selection.

V. EXPERIMENTAL RESULTS AND ANALYSIS

A. Single-Task Baselines vs Cattle-Centered Perception

Table I reports single-task performance comparing raw RGB models against cattle-centered 4-channel configurations on matched evaluation cohorts.

TABLE I
SINGLE-TASK BASELINES (RUNS 1–3) VS CATTLE-CENTERED
PERCEPTION (RUNS 4–6) ON MATCHED EVALUATION COHORTS

Task	Evaluation Metric	RGB	Cattle-Cent.
BCS	Real MAE ↓	0.1929	0.1709
	Exact Acc@0 (%) ↑	40.84	43.57
	Tol. Acc@1 (%) ↑	84.95	89.40
	Macro-F1 ↑	0.4074	0.4039
Behavior	Overall Acc (%) ↑	88.46	87.44
	Balanced Acc (%) ↑	71.30	74.43
	Macro-F1 ↑	0.7378	0.7397
	CVB Surveillance Acc (%) ↑	84.12	80.09
Re-ID (Proto A)	Barn→Parlor R-1 (%) ↑	58.64	63.90
	Barn→Parlor mAP (%) ↑	38.32	40.68
	Snap→Parlor R-1 (%) ↑	38.88	62.93
	Snap→Parlor mAP (%) ↑	27.05	40.42

The single-task evaluations demonstrate that benefits are task-dependent and non-uniform across metrics:

1) *BCS Prediction*: Cattle-centered localization and binary mask guidance reduced physical error from 0.1929 to 0.1709 MAE and increased tolerance accuracy (Acc@1) from 84.95% to 89.40% (+4.45 pp). However, class-balanced metrics showed no corresponding gain: balanced accuracy and Macro-F1 remained flat or slightly lower (0.4074 to 0.4039). Isolating the animal foreground tightened error around true score boundaries without improving discrete class-balanced discrimination.

2) *Behavior Recognition*: Temporal foreground modeling improved balanced accuracy from 71.30% to 74.43% (+3.13 pp), reduced test loss from 0.5505 to 0.4430, and improved minority Walking F1 from 0.2174 to 0.2456. However, overall accuracy declined slightly from 88.46% to 87.44%, and CVB surveillance accuracy dropped from 84.12% to 80.09%, while Kaggle Beef accuracy reached 96.09%.

3) *Cattle Re-ID*: Under oracle ground-truth mask guidance, Barn-to-Parlor Rank-1 improved from 58.64% to 63.90% and mAP from 38.32% to 40.68%. Under extreme camera disparity in Snapshot queries, Rank-1 increased substantially from 38.88% to 62.93% (+24.05 pp) and mAP from 27.05% to 40.42% (+13.37 pp). These gains reflect oracle-guided target framing and cannot be assumed for unvalidated automatic segmentation.

Because localization, cropping, binary masks, and temporal Conv1D were introduced jointly in Runs 4–6, observed gains characterize full model configurations and cannot be attributed specifically to segmentation alone.

B. Multi-Task Learning Performance and Negative Transfer

Table II presents the comparative evaluation across the three multi-task models against the single-task cattle-centered reference (E0).

1) *Negative Transfer in Hard Sharing (E1)*: Monolithic hard sharing achieves a 65.7% parameter reduction (11.9M vs 34.7M across three single-task networks). However, forcing all three tasks through a single shared spatial backbone caused

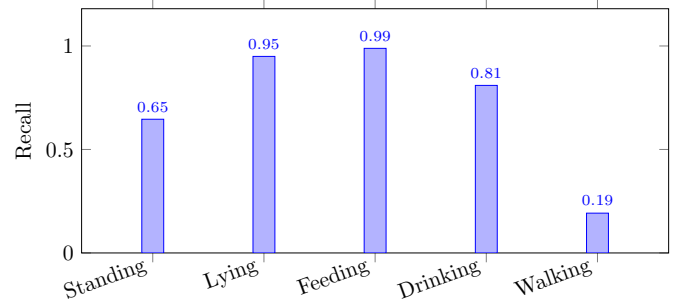


Fig. 5. Single-task behavior recall across 5 action categories on held-out test data. Sedentary postures (feeding, lying) achieve high recall, while walking represents an underrepresented minority class.

outcome-level negative transfer across all three domains: BCS MAE degraded from 0.1709 to 0.1788; Behavior balanced accuracy fell from 74.43% to 67.30% (Walking F1 dropped from 0.2456 to 0.0408); and Re-ID Barn Rank-1 dropped from 63.90% to 57.38% (mAP fell from 40.68% to 30.37%).

2) *Modular Bottleneck Adapters (E3)*: Introducing task-private bottleneck adapters (E3, +3.32% parameters) failed to generally resolve negative transfer. While Behavior test loss (0.4193) and CVB surveillance accuracy (80.33%) improved over E1, BCS MAE degraded further to 0.1916, Re-ID Barn Rank-1 dropped to 49.08% (mAP 28.04%), and Walking F1 collapsed to 0.0000 on the matched held-out test set. Because E3 added 395,904 trainable parameters, performance differences cannot be attributed purely to architectural routing.

3) *Selective Mitigation via PCGrad (E4)*: Gradient projection (E4) operates under exact parameter parity with E1 (11,926,706 parameters). PCGrad provided selective mitigation, particularly on Behavior where overall accuracy reached 86.92% (+1.92 pp over E1), balanced accuracy reached 69.15% (+1.85 pp), Macro-F1 reached 0.7114 (+0.0248), CVB surveillance accuracy reached 81.04% (+4.26 pp over E1), and Walking F1 recovered to 0.0909 (vs. 0.0408 in E1). In BCS, E4 reached 0.1828 MAE, improving upon E3 (0.1916). In Re-ID, E4 recovered Rank-1 over E3 (54.97% vs 49.08% in Barn). However, PCGrad did not eliminate negative transfer: dedicated single-task models remained superior on several primary metrics (BCS MAE 0.1709 vs 0.1828; Behavior balanced accuracy 74.43% vs 69.15%; Walking F1 0.2456 vs 0.0909; Barn Rank-1 63.90% vs 54.97%).

C. Shared-Backbone Gradient Conflict Diagnostics

During 30 epochs of E4 super-step training (16,140 total super-steps), direct diagnostics on the shared 4-channel ResNet-18 backbone recorded **44,177 pairwise conflict projections**, averaging 2.737 projections per super-step. Directional conflict was evenly distributed across task pairings: BCS vs Behavior: 47.8%; BCS vs Re-ID: 46.5%; Behavior vs Re-ID: 47.4%. These diagnostics demonstrate that opposing gradient directions occurred regularly on the shared backbone during E4 training. However, these measurements were recorded specifically under PCGrad projection; they do

TABLE II
HELD-OUT EVALUATION ACROSS SINGLE-TASK REFERENCE (E0) AND MULTI-TASK ARCHITECTURES (E1 HARD-SHARED, E3 MODULAR, E4 PCGRAD)

Task / Domain	Evaluation Metric	Single-Task (E0)	Hard-Shared (E1)	Modular E3	PCGrad (E4)
BCS Evaluation (ScienceDB Test)	Real MAE ↓	0.1709	0.1788	0.1916	0.1828
	Tolerance Acc@1 (%) ↑	89.40	88.44	86.32	87.84
	Exact Acc@0 (%) ↑	43.57	41.10	38.47	40.23
	Macro-F1 ↑	0.4039	0.3605	0.3291	0.3516
Behavior Evaluation (CVB + Beef)	Overall Accuracy (%) ↑	87.44	85.00	85.77	86.92
	Balanced Accuracy (%) ↑	74.43	67.30	66.70	69.15
	Macro-F1 ↑	0.7397	0.6866	0.6755	0.7114
	CVB Surveillance Acc (%) ↑	80.09	76.78	80.33	81.04
	CVB Surveillance F1 ↑	0.6188	0.5402	0.6026	0.6141
	Minority Walking F1 ↑	0.2456	0.0408	0.0000	0.0909
Re-ID Protocol A (Held-Out Cows)	Query Barn → Parlor Rank-1 (%) ↑	63.90	57.38	49.08	54.97
	Query Barn → Parlor mAP (%) ↑	40.68	30.37	28.04	30.23
	Query Snap → Parlor Rank-1 (%) ↑	62.93	57.17	53.71	57.00
	Query Snap → Parlor mAP (%) ↑	40.42	33.69	28.78	33.87
Efficiency	Total Trainable Parameters	34,706,006 (3×)	11,926,706	12,322,610	11,926,706
	Relative Parameter Footprint	100%	34.3%	35.5%	34.3%

not establish that unprojected hard sharing (E1) experienced identical conflict frequencies, nor that gradient conflict was the sole causal mechanism behind E1 degradation.

VI. DISCUSSION, UNCERTAINTY, AND CLAIM BOUNDARIES

1) *Cross-Task Representation Trade-Offs*: Cattle-centered visual representations provide clear utility for individual phenotyping tasks, but benefits depend heavily on task requirements: BCS relies on static body morphology and subcutaneous fat contour; Behavior requires temporal posture transitions and movement dynamics; Re-ID requires fine-grained coat markings. Forcing these divergent objectives onto a single shared ResNet-18 backbone induces outcome-level negative transfer. Neither task-private adapters (E3) nor gradient orthogonalization (E4) uniformly eliminated this interference. Dedicated single-task representations preserved superior performance across primary task criteria. Multi-task consolidation achieves a 65.7% parameter-footprint reduction (11.9M vs 34.7M parameters across independent task models), but deployment involves an engineering trade-off against dedicated single-task accuracy.

2) *Statistical Boundaries and Point Estimates*: All reported single-task and multi-task models were evaluated from single trained checkpoints (point estimates). Repeated-seed training distributions, confidence intervals, and hypothesis tests were not computed. Consequently, reported metric differences represent descriptive point comparisons and are not characterized as statistically significant. While the matched-sample design ensures rigorous internal comparability across identical test subsets (7,549 ScienceDB images, 780 Behavior sequences, 69 held-out Re-ID cows), it does not quantify seed-level or population variance.

3) *Methodological Limitations*: To preserve scientific claim boundaries, key limitations must be highlighted: (i) ScienceDB lacks biological cow IDs; the burst-group-disjoint protocol prevents frame leakage from passage sequences but cannot guarantee biological cow-disjoint evaluation. (ii) Behavior partitions protect video sessions rather than cow IDs; minority Walking occurs exclusively in CVB surveillance (26 test sequences), confounding camera source with activity recognition. (iii) Automated perception retained 93.89% of ScienceDB test images (7,549 / 8,040) and 96.42% of Behavior sequences (780 / 809); perception-failed samples remain outside the evaluated distribution. (iv) Re-ID models used released ground-truth masks; this oracle benchmark does not evaluate automated upstream segmentation. (v) Runs 4–6 combined localization, cropping, binary masks, and temporal convolutions; individual component contributions were not isolated. (vi) E3 added 395,904 parameters (+3.32%), confounding routing with model capacity. (vii) External stress tests remain unexecuted; results do not establish broad generalization to unseen commercial herds. (viii) High keypoint missingness on recumbent cows (72.92% detector failure on lying cattle) and source shortcut risks precluded pose/viewpoint conditioning in primary models.

VII. CONCLUSION AND FUTURE WORK

We evaluated a multi-task deep learning framework unifying Body Condition Scoring, behavior recognition, and biometric cow re-identification under controlled protocols. Cattle-centered visual representations improved BCS error tolerance and boosted open-set Re-ID under extreme camera disparity, though gains were task-dependent and non-uniform across sub-metrics. Naive hard parameter sharing caused outcome-level negative transfer across all three tasks. Task-private

adapters failed to generally alleviate this degradation, while PCGrad provided selective mitigation—notably improving CVB behavior surveillance and recovering minority walking classification—under parameter parity with the hard-shared baseline. Dedicated single-task models remained superior on several primary metrics, showing that multi-task monitoring requires deliberate trade-offs between shared efficiency and task-specific representation requirements.

Future work will focus on: (1) repeated-seed training for confidence intervals; (2) automated segmentation replacing oracle masks in re-identification; (3) multi-herd validation across independent farms; (4) longitudinal biometric stress-testing across lactation and coat growth; (5) factorial ablations isolating perception components; (6) robust cattle pose estimation for recumbent postures; and (7) partial parameter sharing with adaptive loss weighting (e.g., GradNorm [24]) to address cross-task gradient conflict.

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